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Q1: Prove by mathematical induction that

$$\sum_{i=1}^n \frac{1}{i(i+1)} = n/(n+1)$$

Ans → Principle of mathematical induction ⇒

Prove that ⇒ $\sum_{i=1}^n \frac{1}{i(i+1)} = n/(n+1)$

Proof:-

$$\text{L.H.S} = \sum_{i=1}^n \frac{1}{i(i+1)}$$

$$\text{for } n=1 \quad \sum_{i=1}^1 \frac{1}{i(i+1)} = \frac{1}{2}$$

$$\text{for } n=k \quad \sum_{i=1}^k \frac{1}{i(i+1)} = \frac{k}{k+1}$$

$$\text{for } n=k+1 \quad \sum_{i=1}^{k+1} \frac{1}{i(i+1)} = \frac{(k+1)}{(k+1)+1} = \frac{(k+1)}{(k+2)}$$

$$\sum_{i=1}^k \frac{1}{i(i+1)} + \frac{1}{(k+1)(k+2)} = \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k(k+2) + 1}{(k+1)(k+2)} = \frac{k^2 + 2k + 1}{(k+1)(k+2)} = \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

Hence Proved.

Q2: Verify whether $\sqrt{11}$ is rational or irrational.

Ans. The square root of 11, denoted as $\sqrt{11}$, is an irrational number.

To see why, suppose for the sake of contradiction that $\sqrt{11}$ is rational. This means that there exist two integers, say a and b , such that $\sqrt{11} = a/b$, where a and b have no common factors. Squaring both sides, we get:

$$11 = a^2 / b^2$$

Multiplying both sides by b^2 , we get:

$$11b^2 = a^2$$

This means that a^2 is divisible by 11, and hence a must also be divisible by 11 (since 11 is a prime number). Let $a = 11c$ for some integer c . Substituting this into the equation above, we get:

$$11b^2 = (11c)^2 = 121c^2$$

Dividing both sides by 11, we get:

$$b^2 = 11c^2$$

This means that b^2 is also divisible by 11, and hence b must also be divisible by 11. But this contradicts our assumption that a and b have no common factors, since both are divisible by 11. Therefore, our initial assumption that $\sqrt{11}$ is rational must be false, and hence $\sqrt{11}$ is irrational.

Q3: Write the following statements in the symbolic form.

i) Some students can not appear in exam.

ii) Everyone can not sing.

Ans.

(i) Let $S(x)$ be the statement " x is a student" and $E(x)$ be the statement " x can appear in the exam". Then the statement "Some students can not appear in exam" can be written symbolically as:

$$\exists x (S(x) \wedge \neg E(x))$$

This reads as "there exists an x such that x is a student and x cannot appear in the exam".

(ii) Let $P(x)$ be the statement " x is a person" and $S(x)$ be the statement " x can sing". Then the statement "Everyone can not sing" can be written symbolically as:

$$\neg \forall x (P(x) \rightarrow S(x))$$

This reads as "it is not the case that for all x , if x is a person then x can sing". Alternatively, we can also write this statement as:

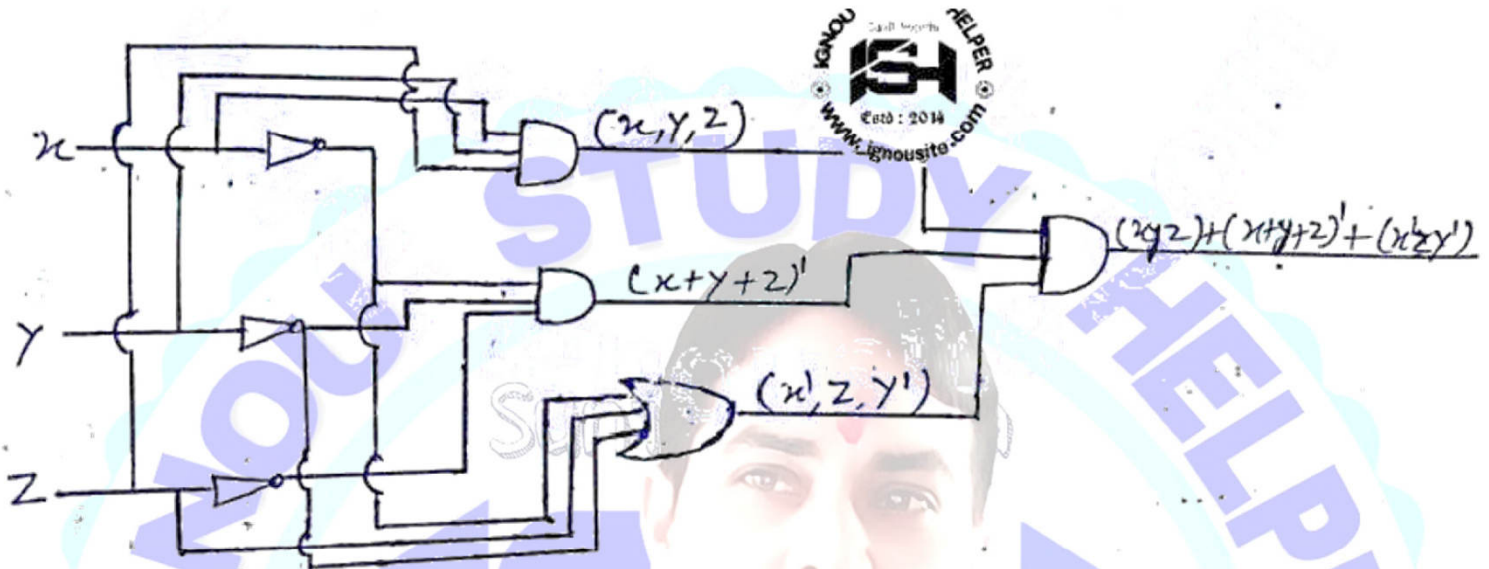
$$\exists x (P(x) \wedge \neg S(x))$$

This reads as "there exists an x such that x is a person and x cannot sing".

Q.4 Draw logic circuit for the following Boolean Expression:

$$(x \cdot y \cdot z) + (x+y+z)' + (x'z'y')$$

Ans. Here is the logic circuit for the given Boolean expression:



Q5: Explain whether function: $f(x) = x^2$ posses an inverse function or not.

Ans.

$$f(x) = x^2$$

$$f(1) = 1^2 = 1$$

$$f(-1) = (-1)^2 = 1$$

So, there is no inverse functions.

$$y = f(x)$$

$$x = f^{-1}(y)$$

$$x = f^{-1}(x^2)$$

$$x = (x^2)^2$$

$$x = x^4$$

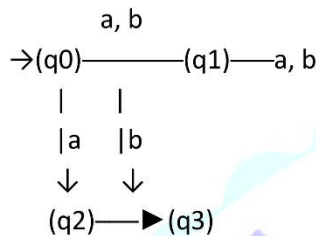
$$x^4 - x = 0$$

So, there is no inverse function.

Q6: Write the finite automata corresponding to the regular expression $(a + b)^*ab$

Ans. The regular expression $(a + b)^*ab$ represents the set of all strings that end with the string "ab".

Here is the finite automaton that corresponds to this regular expression:



In this automaton, q_0 is the start state, and q_3 is the accepting state. The arrows represent transitions on input symbols.

From state q_0 , the automaton can transition to either q_0 or q_1 on input symbol "a" or "b". From q_1 , the automaton can only transition to q_2 on input symbol "a" and to q_3 on input symbol "b". Finally, from q_2 , the automaton can only transition to q_3 on input symbol "b".

To recognize a string using this automaton, we start at the start state q_0 and read the input symbols one by one. If we reach the end of the string and we are in the accepting state q_3 , then the string is accepted by the automaton and is in the language represented by the regular expression $(a + b)^*ab$.

Q7: If L_1 and L_2 are context free languages then, prove that $L_1 \cup L_2$ is a context free language.

Ans. To prove that $L_1 \cup L_2$ is a context-free language, we can construct a context-free grammar that generates the language.

Let $G_1 = (V_1, \Sigma, R_1, S_1)$ be a context-free grammar for L_1 and $G_2 = (V_2, \Sigma, R_2, S_2)$ be a context-free grammar for L_2 .

We can construct a new grammar $G = (V, \Sigma, R, S)$ for $L_1 \cup L_2$ by introducing a new start symbol S and adding the productions:

$$S \rightarrow S_1 \mid S_2$$

The non-terminal symbols V of the new grammar G are the union of the non-terminal symbols of G_1 and G_2 , i.e. $V = V_1 \cup V_2 \cup \{S\}$. The set of terminals Σ is the same for both languages L_1 and L_2 , and R is the union of R_1 and R_2 together with the new production $S \rightarrow S_1 \mid S_2$.

The production $S \rightarrow S_1$ generates all strings in L_1 , and $S \rightarrow S_2$ generates all strings in L_2 . Since G_1 and G_2 are context-free, the productions in R_1 and R_2 are context-free, and therefore the productions in R are also context-free.

Thus, we have shown that $L_1 \cup L_2$ is a context-free language.

Q8: Explain Decidable and Undecidable Problems. Give example for each.

Ans. Decidable Problem: A decision problem is decidable if there exists an algorithm that can solve it in a finite amount of time for any given input. In other words, a decidable problem is one for which there is a definitive and algorithmic way to determine the answer.

Example: Given a list of numbers, is there a pair of numbers in the list whose sum is equal to a specific value? This is a decidable problem because we can solve it using a straightforward algorithm. We can iterate over all possible pairs of numbers in the list

and check whether their sum equals the specific value. If we find such a pair, we can return "yes"; otherwise, we can return "no".

Undecidable Problem: An undecidable problem is one for which there is no algorithm that can solve it for all possible inputs. That is, there is no finite algorithm that can always provide a definite yes or no answer to the problem.

Example: The halting problem is a classic example of an undecidable problem. Given a program and an input, the halting problem asks whether the program will eventually halt (i.e., stop running) when given that input. It has been proven that there is no algorithm that can solve the halting problem for all possible programs and inputs. This means that there are some programs for which it is impossible to determine whether they will halt or run indefinitely.

Q9: What is equivalence relation? Explain use of equivalence relation with the help of an example.

Ans. Equivalence relation: An equivalence relation is a relation between elements of a set that satisfies three conditions: reflexivity, symmetry, and transitivity. That is, if we have a set S and a relation R on S , then R is an equivalence relation if and only if:

1. Reflexivity: for any element a in S , aRa (that is, a is related to itself).
2. Symmetry: for any elements a and b in S , if aRb , then bRa (that is, if a is related to b , then b is related to a).
3. Transitivity: for any elements a , b , and c in S , if aRb and bRc , then aRc (that is, if a is related to b and b is related to c , then a is related to c).

Use of equivalence relation: Equivalence relations are useful in many areas of mathematics and computer science, including group theory, graph theory, and relational databases.

Example: One example of an equivalence relation is the relation of congruence modulo n on the set of integers $\mathbb{Z}$. For any integers a and b , we say that a is congruent to b modulo n (written $a \equiv b \pmod{n}$) if and only if $a - b$ is divisible by n . This relation is an equivalence relation because it satisfies the three conditions:

1. Reflexivity: for any integer a , $a \equiv a \pmod{n}$, because $a - a = 0$ is divisible by n .
2. Symmetry: if $a \equiv b \pmod{n}$, then $a - b$ is divisible by n . But if $a - b$ is divisible by n , then so is $-(a - b) = b - a$, so $b \equiv a \pmod{n}$.
3. Transitivity: if $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a - b$ is divisible by n and $b - c$ is divisible by n . Therefore, $(a - b) + (b - c) = a - c$ is divisible by n , so $a \equiv c \pmod{n}$.

This relation is useful in many applications, including cryptography and number theory.

Q10: There are three Companies, C1, C2 and C3. The party C1 has 4 members, C2 has 5 members and C3 has 6 members in an assembly. Suppose we want to select two persons, both from the same Company, to become president and vice president. In how many ways can this be done?

Ans. We need to select two persons from the same company and then select one as president and the other as vice president. There are three companies, and for each company, we need to choose 2 persons, and then we need to select one person as president and the other as vice president. Thus, the total number of ways is:

$$3 * C(4, 2) * 2! + 3 * C(5, 2) * 2! + 3 * C(6, 2) * 2!$$

where $C(n, k)$ is the number of ways to choose k items from a set of n items.

Evaluating this expression, we get:

$$3 * C(4, 2) * 2! + 3 * C(5, 2) * 2! + 3 * C(6, 2) * 2!$$

$$= 3 * 6 * 2 + 3 * 10 * 2 + 3 * 15 * 2$$

$$= 36 + 60 + 90$$

$$= 186$$

Therefore, there are 186 ways to select two persons, both from the same Company, to become president and vice president.

Q11: How many words can be formed using letter of DEPARTMENT using each letter at most once?

i) If each letter must be used,

ii) If some or all the letters may be omitted.

Ans.

i) If each letter must be used,

Ans. In the word "DEPARTMENT", total number of letters = 10 among which 'E' appears 2 times, 'T' appears 2 times and other than 'E' and 'T' all letters appearing 1 time each.

So the number of distinct letters = $(10 - 2) = 7$ (D, E, P, A, R, T, M, N)

The number of words can be formed using all the distinct letters of the word "DEPARTMENT" is given by –

$$= 8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40,320$$

ii) If some or all the letters may be omitted.

Ans. If some or all the letters may be omitted, we can form words of different lengths. We can form words of length 1, 2, 3, ..., up to 10 by selecting a subset of the letters in DEPARTMENT and arranging them in all possible orders.

For a word of length k, we need to select k letters from DEPARTMENT and arrange them in k! ways. The total number of words that can be formed is the sum of the number of words of length 1, 2, 3, ..., up to 10.

Number of words of length 1 = 10

Number of words of length 2 = $C(10, 2) \times 2! = 45 \times 2 = 90$

Number of words of length 3 = $C(10, 3) \times 3! = 120 \times 6 = 720$

Number of words of length 4 = $C(10, 4) \times 4! = 210 \times 24 = 5,040$

Number of words of length 5 = $C(10, 5) \times 5! = 252 \times 120 = 30,240$

Number of words of length 6 = $C(10, 6) \times 6! = 210 \times 720 = 151,200$

Number of words of length 7 = $C(10, 7) \times 7! = 120 \times 5,040 = 604,800$

Number of words of length 8 = $C(10, 8) \times 8! = 45 \times 40,320 = 1,814,400$

Number of words of length 9 = $C(10, 9) \times 9! = 10 \times 362,880 = 3,628,800$

Number of words of length 10 = $C(10, 10) \times 10! = 1 \times 3,628,800 = 3,628,800$

The total number of words that can be formed by using some or all the letters in DEPARTMENT is the sum of the above numbers, which is:

$$10 + 90 + 720 + 5,040 + 30,240 + 151,200 + 604,800 + 1,814,400 + 3,628,800 + 3,628,800 = 10,393,980$$

Therefore, there are 10,393,980 words that can be formed using the letters of DEPARTMENT, where each letter may be used at most once.

Q12: What is the probability that a number between 1 and 10,000 is divisible by neither 2, 3, 5 nor 7?

Ans. Let N be the total number of integers between 1 and 10,000, inclusive. We can count the integers between 1 and 10,000 that are divisible by 2, 3, 5, or 7 and use the principle of inclusion-exclusion to find the number of integers that are divisible by at least one of these numbers. Then, we can subtract this count from N to get the number of integers that are not divisible by any of them. Finally, we divide this number by N to get the probability that a number between 1 and 10,000 is not divisible by 2, 3, 5, or 7.

There are 5,000 even integers between 1 and 10,000. There are 3,333 integers divisible by 3, 2,000 integers divisible by 5, and 1,428 integers divisible by 7. We need to subtract the integers that are divisible by at least two of these numbers, then add back the integers that are divisible by at least three of them, and finally subtract the integers that are divisible by all four of them. To do this, we use the principle of inclusion-exclusion:

Number of integers divisible by at least one of 2, 3, 5, or 7 = $5000 + 3333 + 2000 + 1428 - (1666 + 1000 + 714 + 285) + (333 + 142 + 95) - 47 = 7680$

Therefore, the number of integers between 1 and 10,000 that are not divisible by 2, 3, 5, or 7 is:

$$N - 7680 = 10,000 - 7680 = 2,320$$

The probability that a number between 1 and 10,000 is not divisible by 2, 3, 5, or 7 is:

$$P(\text{not divisible by 2, 3, 5, or 7}) = 2320 / 10000 = 0.232$$

Thus, the probability that a number between 1 and 10,000 is not divisible by 2, 3, 5, or 7 is 0.232.

Q13: Explain inclusion-exclusion principle and Pigeon Hole Principle with example.

Ans. Inclusion-Exclusion Principle: The inclusion-exclusion principle is a counting technique used to find the size of the union of two or more sets. It states that the size of the union of two or more sets is equal to the sum of their sizes, minus the size of their intersection(s), plus the size of the intersection(s) of their pairwise combinations, minus the size of the intersection(s) of their triple-wise combinations, and so on. The formula for the inclusion-exclusion principle is as follows:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

and so on for any number of sets.

Example: Suppose we have three sets A , B , and C , where $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7\}$. We want to find the size of their union.

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C| = 4 + 4 + 4 - 2 - 1 - 1 + 1 = 9$$

Therefore, the size of the union of A , B , and C is 9.

Pigeonhole Principle: The pigeonhole principle is a basic principle in combinatorics that states that if n items are placed into m containers where $n > m$, then at least one container must contain more than one item. The principle is often used to show that a certain situation or outcome is inevitable or to find lower bounds on the number of objects that must be used to satisfy a certain property.

Example: Suppose we have 10 apples and 9 baskets. We want to put the apples into the baskets such that each basket has at most one apple. Since we have more apples than baskets, by the pigeonhole principle, there must be at least one basket that has more than one apple.

Q14: Find an explicit recurrence relation for minimum number of moves in which the n -disks in tower of Hanoi puzzle can be solved! Also solve the obtained recurrence relation through an iterative method.

Ans. The minimum number of moves required to solve the Tower of Hanoi puzzle with n disks can be obtained recursively as follows:

Let $M(n)$ denote the minimum number of moves required to solve the puzzle with n disks.

- If $n=1$, it can be solved in 1 move.
- If $n>1$, then we need to move the top $(n-1)$ disks to a spare peg, move the n th disk to the target peg, and then move the $(n-1)$ disks from the spare peg to the target peg. This requires $2M(n-1) + 1$ moves.

Therefore, we have the following recurrence relation for the minimum number of moves:

$$M(n) = 1 \text{ if } n=1 \quad M(n) = 2M(n-1) + 1 \text{ if } n>1$$

Using this recurrence relation, we can solve for $M(n)$ iteratively as follows:

$$M(1) = 1$$

$$M(2) = 2M(1) + 1 = 3$$

$$M(3) = 2M(2) + 1 = 7$$

$$M(4) = 2M(3) + 1 = 15$$

$$M(5) = 2M(4) + 1 = 31 \dots$$

In general, we can obtain $M(n)$ by doubling the previous value of M and adding 1. Therefore, the explicit formula for $M(n)$ is:

$$M(n) = 2^n - 1$$

So, the minimum number of moves required to solve the Tower of Hanoi puzzle with n disks is $2^n - 1$.

Q.15 Find the solution of the recurrences relation $a_n = a_{n-1} + 2a_{n-1}$, $n > 2$ with $a_0 = 0$, $a_1=1$

Ans.

Ans: Recurrences relation $\Rightarrow$

$$a_n = a_{n-1} + 2a_{n-2}$$

$n \geq 2$ with $a_0 = 0$

$a_1 = 1$

$$a_n - a_{n-1} - 2a_{n-2} = 0$$

is second order linear homogeneous recurrence with constant coefficient.

Let $a_n = r^n$ is a solution

$$r^2 - r - 2 = 0$$

$$(r-2)(r+1) = 0$$

or $r = 2, -1$ distinct roots.

$$a_n = b_1 (2)^n + b_2 (-1)^n$$

Again $a_0 = 0 \Rightarrow b_1 + b_2 = 0$

$a_1 = 1 \Rightarrow 2b_1 - b_2 = 1$

The solution of these two equations are

$b_1 = \frac{1}{3}$ and $b_2 = -\frac{1}{3}$

Hence the equation is given by

$$a_n = \left(\frac{1}{3}\right) 2^n - \left(\frac{1}{3}\right) (-1)^n$$

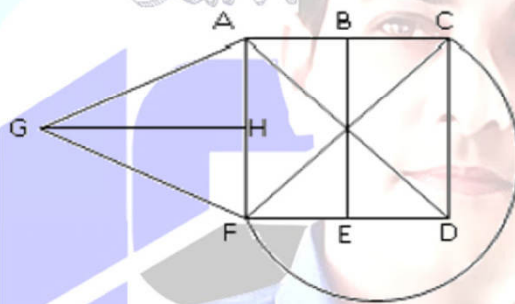
Q.16 Prove that the complement of G is $\bar{G}$.

Ans. The complement of a graph G is another graph that has the same set of vertices as G , but has edges between any two vertices in G that are not adjacent in G , and does not have edges between any two vertices in G that are adjacent in G . To prove that the complement of G is $\bar{G}$, we need to show that every vertex in G is adjacent to every vertex that is not adjacent to it in G , and that no vertex in G is adjacent to any vertex that is adjacent to it in G .

Let u and v be two vertices in G that are not adjacent in G . Then in the complement of G , there is an edge between u and v , since they are not adjacent in G . But this means that in G , u and v are not connected by an edge. Similarly, if u and v are adjacent in G , then in the complement of G , there is no edge between u and v , since they are adjacent in G . This means that in $\bar{G}$, u and v are connected by an edge.

Since this holds for every pair of vertices in G , we can conclude that the complement of G is $\bar{G}$.

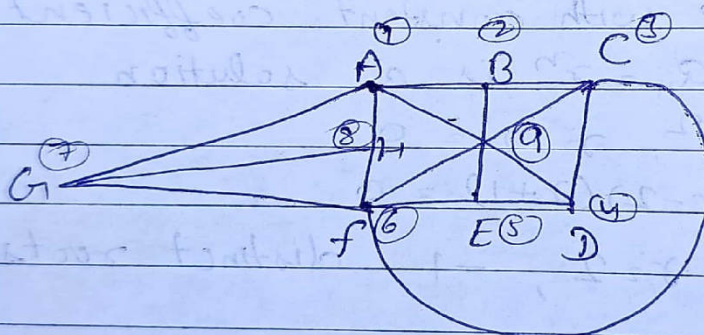
Q17: What is a chromatic number of a graph? What is a chromatic number of the following graph?



Ans.

Chromatic graph $\rightarrow$ Minimum number of colours required to colour the given graph. Chromatic graph made a coloured the graph for colouring the graph.

The chromatic number of the graph $\rightarrow$



$$\chi(G) = 9$$

Q18: Determine whether the above graph has a Hamiltonian circuit. If it has, find such a circuit. If it does not have, justify it.

Ans.

Q. 18

It has not Hamilton circuit.

denoted by $\chi(G)$ or k -chromatic graph

It is defined as the least number of colours needed for colouring the graph

Hamilton circuit is a cycle in an undirected graph that visits each vertex exactly once and returns to the starting vertex.

Hence Proved.

Q19: Explain and prove the Handshaking Theorem, with suitable example.

Ans. Handshaking Theorem: The Handshaking Theorem, also known as the Handshaking Lemma or the Handshake Lemma, is a result in graph theory that provides a relationship between the number of edges and the degrees of vertices in an undirected graph. The theorem states that the sum of the degrees of all vertices in an undirected graph is twice the number of edges.

Formally, the Handshaking Theorem can be stated as follows:

Let $G = (V, E)$ be an undirected graph, where V is the set of vertices and E is the set of edges. Let $d(v)$ denote the degree of a vertex v , i.e., the number of edges incident to v . Then, we have:

$$\sum(d(v)) = 2 * |E|$$

where the sum is taken over all vertices v in V , and $|E|$ denotes the cardinality (i.e., number of elements) of the set E .

Proof: To prove the Handshaking Theorem, we can count the total number of endpoints of edges in the graph in two different ways: by summing the degrees of all vertices, and by counting the endpoints of edges.

Let's consider an undirected graph $G = (V, E)$, where V is the set of vertices and E is the set of edges. For any vertex v in V , let $d(v)$ denote the degree of v , i.e., the number of edges incident to v . Then, the total number of endpoints of edges in G is equal to the sum of the degrees of all vertices, i.e.,

$$\sum(d(v)).$$

On the other hand, each edge in E has two endpoints, so the total number of endpoints of edges in G is equal to twice the number of edges, i.e.,

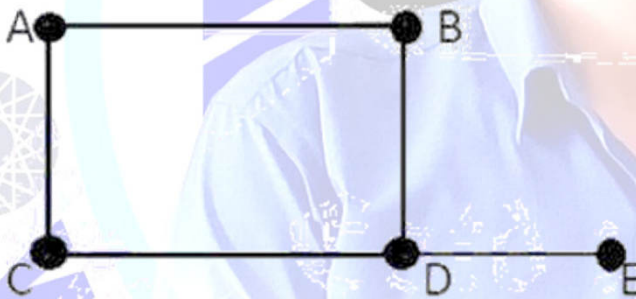
$$2 * |E|.$$

Therefore, we have:

$$\sum(d(v)) = 2 * |E|$$

which proves the Handshaking Theorem.

Example: Let's consider the following undirected graph G :



The graph has 5 vertices and 6 edges. We can calculate the degrees of each vertex as follows:

- $d(A) = 2$ (A is incident to edges AB and AC)
- $d(B) = 2$ (B is incident to edges AB and BD)
- $d(C) = 2$ (C is incident to edges AC and CD)
- $d(D) = 3$ (D is incident to edges BD, CD, and DE)
- $d(E) = 1$ (E is incident to edge DE)

Therefore, the sum of the degrees of all vertices is:

$$\sum(d(v)) = 2 + 2 + 2 + 3 + 1 = 10$$

On the other hand, the number of edges is:

$$|E| = 6$$

Using the Handshaking Theorem, we can verify that:

$$\text{sum}(d(v)) = 2 * |E| \quad 10 = 2 * 5$$

which holds true.

Q20: Explain the terms PATH, CIRCUIT and CYCLES in context of Graphs.

Ans. In graph theory, a "graph" is a mathematical structure that consists of a set of vertices (also known as nodes) and a set of edges that connect them.

1. **PATH:** A path in a graph is a sequence of vertices connected by edges. In other words, a path is a route through the graph from one vertex to another. The length of a path is the number of edges it contains.
2. **CIRCUIT:** A circuit (also called a cycle) in a graph is a closed path that starts and ends at the same vertex. In other words, a circuit is a path that returns to its starting vertex without repeating any other vertices. The length of a circuit is the number of edges it contains.
3. **CYCLES:** A cycle is a subset of edges in a graph that form a closed path. In other words, it is a sequence of edges that connect a sequence of vertices such that the first and last vertices are the same. Cycles can be of any length, from 3 or more vertices to infinitely long paths in the case of infinite graphs.